



Elsevier • Book proposal form

Working title

Titles and subtitles should be focused to include key terms that readers would use if searching for information on this topic

Two related but independently publishable monographs are proposed, to be considered together. Publishing preference: as a matched pair — a two-volume research programme under one publishing agreement — with each volume deliberately self-contained so that individual publication of either one remains fully viable (individual consideration is acceptable should the imprint prefer to scope only one project).

Volume 1 (foundations): Effective Causal Descent: Uniform Selection, Contextual Gluing, and Causal Histories. A research monograph on decision-making under partial observation, developing a section-theoretic framework in which local validity, compatibility, effectivity, and resource bounds are four presentation-relative diagnostics that pinpoint exactly where the local-to-global assembly of decision rules fails — across constraint satisfaction, sheaf contextuality, Bell nonlocality, computable analysis, and extensive-form games.

Volume 2 (algorithms): Algorithms and Certificates for Finite Effective Causal Descent: Width-Based Synthesis for Sensor-Limited Mechanical Control. The algorithmic companion: a proof-producing synthesis pipeline for finite sensor-limited mechanical models, with typed certificates, tree-decomposition dynamic programming, a certified mechanics front end, a complete design-side complexity classification, and a reproducible public software artifact.

The subtitles deliberately split the searchable key terms across the two books (constraint satisfaction, sheaf semantics, contextuality, Bell nonlocality, Weihrauch degrees, extensive-form games in Volume 1; certifying algorithms, tree decompositions, finite-element certification, sensor placement in Volume 2).

Keywords

Include key terms (not already included in the title/subtitle) that readers would use if searching for information on this topic

Volume 1: partial observability; decision-making under incomplete information; presentation-relative diagnostics; constraint satisfaction; sheaf semantics and contextuality; Bell nonlocality and repair; computable selection; Weihrauch reducibility; extensive-form games; causal models and causal protocols.

Volume 2: certifying algorithms; typed certificates; tree decompositions and treewidth; weighted width; parameterized complexity; NP-completeness and $W[2]$ -hardness; sensor placement and sensor repair;

finite-element certification; sparsification; reproducible research software.

Author/Editor information

- Specify if this book is authored or edited
- Include address and contact details, qualifications and experience, and a short biography for all book authors and/or editors

Both volumes are authored (single-author research monographs), not edited.

To be completed by the author before submission:

[Full legal name]; [position / independent researcher]; [institutional or postal address]; [email]; [telephone]; [ORCID].

Qualifications: [degrees, field, year].

Short biography (draft for the author to adapt): The author works on the mathematics of decision-making under partial observation and on certifying algorithms for engineering models. The present pair of monographs grew out of a multi-year research programme combining constraint-satisfaction theory, sheaf contextuality, computable analysis, and certified computation; the complete programme — both manuscripts, the implementation reports, and the software artifact — is version-controlled and publicly archived.

Primary audience

Please indicate the **most relevant target audience**

Researchers and graduate students in theoretical computer science — specifically the constraint-satisfaction, structural-decomposition (treewidth), and parameterized-complexity communities — who need a unified account of when local consistency guarantees global solutions, when it provably does not, and how an algorithm can certify its own answers in either direction.

Secondary audience

Please describe your entire audience in as much detail as possible, e.g. industry sector, job role, level, subject specialism. If the book could be used for a course please provide details, including program and level.

The audience spans five research communities and two levels:

- AI and decision theory: researchers on planning and control with limited sensing (adjacent to POMDPs, but relational and exact rather than probabilistic).
- Formal methods and verified computing: researchers interested in certificate-producing algorithms and independent verification of solver output (SAT/CP/MILP cross-checking).
- Quantum foundations: researchers in sheaf-theoretic contextuality and Bell nonlocality (Volume 1 supplies the decision-theoretic side and the repair theory absent from that literature).
- Computable analysis and Weihrauch complexity: researchers working on effectivity tiers (Volume 1's computability chapter gives new examples: DNR classes, oracle spectra, WKL degrees).

– Mechanical, civil, and control engineering: practitioners of sensor placement, structural health monitoring, and digital-twin modelling (Volume 2's certified front end and benchmark suite).

Course use: Volume 1 can support a 12–15 week advanced graduate seminar in mathematics or computer science on "local-to-global methods" (constraint satisfaction, contextuality, computability). Volume 2 can support a graduate module on certifying algorithms and structural decomposition methods, with the software artifact as a reproducibility lab. Level: MSc/PhD in computer science, applied mathematics, or control engineering.

Background and purpose

- Provide a brief description of this book, similar to what you would find on the back cover of the published book
- What is your purpose in writing this book? Why is there a need for a new resource in this area?
- What **problem** does this book solve for readers?

In plain, everyday terms — what this work is for. Every person, and every machine, acts on an incomplete picture of the world. We choose what to do next without being able to see, measure, or distinguish everything that matters — and when a plan fails, the honest question is whether the world defeated us, or whether our own blind spots did. These two monographs are about that situation, made precise. They ask a single question: when an agent cannot tell certain possibilities apart, what can it still achieve, where exactly does its power end, and how can that boundary be proved rather than guessed? The desire the books serve is the ordinary human one — to understand the world, and to understand the limits of what any observer inside that world can know and control. What is new is that the answer is diagnosable: the theory names the exact step at which acting on partial information breaks, and the software certifies, for concrete engineering models, both what is possible and what is provably impossible at a given sensing budget.

Volume 1 (back-cover description). From a single constraint — an interface cannot support different responses to worlds it does not distinguish — the book derives four presentation-relative diagnostics: local validity, compatibility of a declared diagram, effective realisability, and bounded realisability. The labels are not invariants of bare capability; they record where success first fails in a chosen local-to-global presentation. Compactness, Helly, join-tree, parity, support-contextuality, source-integration, and Bell-coupling results provide finite certificate and gluing tools; an obstruction datum records objectwise local success together with the nested global solution spaces. A full morphism theory (functoriality, composition laws, guarded feedback) follows, and the framework is interfaced with four literatures usually treated separately: constraint satisfaction and sheaf contextuality; Bell nonlocality (Farkas certificates, finite-sample testing, repair profiles); computable analysis (Weihrauch degrees, DNR classes, oracle spectra); and extensive-form games and causal protocols.

Volume 2 (back-cover description). The algorithmic companion. A

certificate-producing dynamic program over tree decompositions, with ordinary and weighted-width bounds, emits typed, independently checkable certificates: policies, refutations, optimum costs, fibre conflicts, sensor repairs, finite-element rows, and sparsification ladders. A mechanics front end converts declared truss and mass–spring models with rational data into certified exact influence rows and brackets the true policy space in a proved inner/outer sandwich; width-adaptive sparsification gives monotone certified refinement. A recorded computational study benchmarks the pipeline against SAT, CP-SAT, and MILP solvers on identical instances, accepts an external answer only when cross-checked by an independently verified native certificate, and is version-exactly replicated from a public artifact. The design side of the problem is completely classified: NP-completeness, W[2]-hardness, logarithmic approximability with a matching barrier under $P \neq NP$, and ETH-tight limits.

Purpose and need. The problem the pair solves for readers is fragmentation plus unverifiability. Local-to-global failure under partial information is studied in five nearly disjoint literatures (CSP, contextuality, games, computable analysis, sensor-limited control), each with its own vocabulary and none with a shared diagnostic language; and the algorithmic practice in this space trusts solver output without independent proof. No existing resource gives a reader both the unified theory of where partial-information decision-making fails and a certificate-backed, reproducible toolchain for the finite case. The intent, plainly stated, is to replace "it seems to work" with "here is the proof, and here is exactly where the limits of what can be known and controlled lie" — the mathematics of acting well within one's own limits, and of knowing, provably, where those limits are.

Publishing preference (per your guidance note). The projects are proposed as a matched pair — two volumes of one research programme with shared notation and mutual cross-references, contracted together — as the primary preference. Each volume is nevertheless fully self-contained (all notation defined, all proofs included, cross-references non-load-bearing), so individual consideration of either volume is acceptable if the imprint prefers to scope only one project.

Benefits to audience

With reference to the target audience(s) listed above, please give details of:

- The information needs and daily challenges of the audience relating to the subject of the book and how your book will address these needs, challenges or pain points
- List **three key features** and content in your book that will be most valuable to the reader

The audience's recurring pain points, and how the pair addresses them:

- Fragmentation. A constraint-satisfaction researcher who wants the contextuality or computability analogue of a local-consistency result currently re-derives it across four vocabularies. The pair gives one framework and one notation, with the standard results recast as instances, and a diagnostic label (L, D, K, R) that transfers across all five communities.
- Unverifiable answers. Practitioners of sensor-limited control accept SAT/CP/MILP output on trust. The books provide a typed certificate grammar so that any answer — including "no solution exists at this budget", which raw solver output cannot establish — can be verified by a small independent checker.
- Irreproducible computational claims. Published benchmark studies are routinely unreproducible. The artifact discipline (version-frozen tag, recorded environment, one-command replication) is itself a template readers can adopt.

Three key features most valuable to the reader:

1. The four-diagnostic framework and obstruction datum (Volume 1). A reader can name exactly where a local-to-global construction fails — locally impossible, incompatible, uncomputable, or too costly — and transport that diagnosis across constraint satisfaction, contextuality, games, and computability, with the nested global solution spaces made explicit.

2. Certificate-producing width-bounded synthesis (Volume 2). Tree-decomposition dynamic programming with independently checkable message tables and typed certificates (policies, refutations, optima, fibre conflicts, sensor repairs, FE rows, sparsification ladders), together with the complete design-side complexity classification telling the reader when not to search for an efficient design method at all.

3. The reproducible artifact. A public, version-frozen prototype that generates and verifies every certificate type, replays the recorded benchmark and solver-comparison studies with one command, and rejects deliberately corrupted certificates — the computational chapters can be audited, not merely read.

Inclusion and reaching a global audience

Elsevier serves a wide and global audience in the English language. Please outline planned features that will help your

– Single-authored research monographs, but the programme integrates and cites the international research literature across five communities (constraint processing, parameterized complexity, quantum contextuality, computable analysis, structural engineering), so readers from each research culture find a familiar entry point and a complete

project reach a wide audience, for example:

- A broad range of qualified international authors and contributors, or case studies
- Contributions from leading figures, including subject matter experts at all stages of their careers
- Experience and insight from authors or contributors in academia, industry, or national labs.

Elsevier is committed to providing accessible content and increasing inclusion and equity in research, healthcare and education publishing. Should your project move forward, you will be asked to follow our [Inclusion and Accessibility guidelines](#).

bibliography.

– The software artifact is public and dependency-free in its core (pure Python; external solvers optional and clearly separated), and runs on modest hardware — the recorded experiments used a 2-core, 4 GiB virtual machine. Students and researchers at under-resourced institutions can reproduce every computational claim without licenses or compute clusters.

– Both volumes are written as self-contained exposition: all notation is defined at first use, a shared notation table opens each book, and the English is deliberately plain — supporting self-study by early-career researchers and non-native speakers.

– The author will follow Elsevier's Inclusion and Accessibility guidelines for the final files (accessible tables, alternative text for figures, colour-independent figure design).

Competition

Please list the books, websites or other information sources that would compete most closely with your book and briefly describe how they compare and why readers would choose your book over the competition

No existing title competes with the pair as a whole; the nearest titles are per-community, which is itself the market gap the pair fills.

Nearest titles for Volume 1:

– R. Dechter, Constraint Processing (Morgan Kaufmann/Elsevier, 2003) — the standard CSP reference for local consistency and structural decomposition; Volume 1 differs by making failure presentation-relative and by unifying CSP with contextuality, computability, and games.

– K. R. Apt, Principles of Constraint Programming (Cambridge University Press, 2003) — broader CSP theory, same community boundary.

– The sheaf-contextuality framework (Abramsky and Brandenburger, "The sheaf-theoretic structure of non-locality and contextuality", Communications in Mathematical Physics, 2011, and its successors) exists only as journal articles; to our knowledge no dedicated research monograph exists. Volume 1 supplies the first book-length, decision-theoretic treatment, including repair profiles absent from that literature.

– K. Weihrauch, Computable Analysis (Springer, 2000) and the Elsevier Handbook of Computability and Complexity in Analysis (Brattka and Hertling, eds, 2021) — the reference works for computable analysis; Volume 1's effectivity tier is complementary, supplying the selection-and-decision side those volumes do not treat.

– M. Osborne and A. Rubinstein, A Course in Game Theory (MIT Press,

1994) — extensive-form games without partial-observation diagnostics or a computability tier.

Nearest titles for Volume 2:

– T. Klops, *Treewidth: Computations and Approximations* (Springer, Lecture Notes in Computer Science 842, 1994) — treewidth computations on graphs; Volume 2 supplies the certifying, application-level pipeline (typed certificates, mechanics front end, budgeted repair) that Klops does not attempt.

– Cygan et al., *Parameterized Algorithms* (Springer, 2015) — the FPT textbook; Volume 2's certifying width-DP is its decision-and-verification-oriented counterpart with explicit certificates and an engineering front end.

– D. Koller and N. Friedman, *Probabilistic Graphical Models* (MIT Press, 2009) — junction-tree inference for probabilistic models; Volume 2 treats relational (exact, non-probabilistic) constraints with proofs of unsatisfiability and cost optimality, which probabilistic inference does not certify.

– C. R. Farrar and K. Worden, *Structural Health Monitoring: A Machine Learning Perspective* (Wiley, 2013) — the engineering side of sensing; statistical and data-driven, with no exact synthesis or refutation certificates. Volume 2's certified-exact approach is complementary rather than overlapping.

– K. Mehlhorn and P. Sanders, *Algorithms and Data Structures: The Basic Toolbox* (Springer, 2008) — introduces certifying algorithms as a discipline; Volume 2 is the book-length application of that discipline to decision problems under partial observation.

Why readers choose these books: no competing title offers (i) the unified local-to-global diagnostic framework spanning five communities, (ii) certificate-backed algorithms whose every answer — positive or negative — is independently checkable, or (iii) a fully reproducible public artifact behind the computational study. Each competitor covers one slice; the pair covers the whole vertical from foundations to checkable engineering answers.

Table of contents

Please provide here or attach separately the planned contents of your book, including chapter titles and a sentence or two related to the scope/initial content plan for each chapter

Volume 1 — Effective Causal Descent: Uniform Selection, Contextual Gluing, and Causal Histories (14 chapters plus appendix):

1. Introduction — motivation, the observational-uniformity constraint, the four diagnostics, chapter map. 2. Minimal selector kernel — the fibrewise selection lemma at the static core. 3. Prediction-game

If your book is edited, please include a tentative list of the chapter authors (they do not need to be finalized at this time)

instantiation — the framework read as a prediction game. 4. Static local obstructions and past factorisation — the local-failure diagnostic and past-history factorisation. 5. Context diagrams and effective descent — context categories, restriction arrows, descent. 6. Well-formedness, semantic outcomes, and transport — the semantic ladder transporting diagnostics across presentations. 7. Distributed selectors and canonical compatibility presentations — the compatibility diagnostic in distributed form. 8. Quantitative decomposition — quantitative refinements of the decomposition tools. 9. Finite gluing, contextuality, and coupling — the finite tier: compactness, Helly, join-tree and parity tools, support contextuality, Bell coupling. 10. Effectivity, computability, and resources — the effectivity diagnostic: DNR classes, oracle spectra, WKL Weihrauch degrees, capability-index completeness, resource bounds. 11. Extensive-form causal descent — backward induction, active-sensing relativity, confluent causal-poset execution, DNR/PA transfer. 12. Functorial and compositional effective causal descent — the morphism theory, composition laws, guarded feedback. 13. Conjectural extensions — open problems with their exact status. 14. Conclusion. Appendix: domain-theoretic feedback.

Volume 2 — Algorithms and Certificates for Finite Effective Causal Descent (14 sections plus two appendices):

1. Introduction — scope, contributions, relation to prior work. 2. Finite ECD presentations — the canonical finite input format. 3. Certificate grammar and status discipline — typed certificates and the inconclusive-status rule. 4. One-shot fibres and causal compilation — the fibre criterion. 5. Finite complexity boundary — the classification table, including the design-side NP-completeness and $W[2]$ -hardness results. 6. Width-based certifying dynamic programming — the DP, its soundness and bounds, the fixed-parameter reading. 7. Sensor repair — the repair theorem and the design-side classification (listed-candidate and unrestricted synthesis). 8. Certified affine mechanics front end — structural models to certified influence rows; the sound inner/outer sandwich. 9. Width-adaptive sparsification — monotone certified refinement ladders. 10. Exact demonstrator — the two-spring example and one repairing sensor. 11. Mechanics benchmark suite — truss and mass-spring families with quantised sensors. 12. Width-adaptive sparsification study — retained-scope ladders. 13. Prototype artifact and reproducibility — the recorded studies and version-exact replication. 14. Limitations and conclusions — all boundaries stated explicitly. Appendix A: standard reductions for the explicit-table frontier. Appendix B: the artifact appendix (formats, reproduction commands, claim boundary).

Publishing timeline

Complete manuscript: both volumes are complete and compiled now (Volume 1: 101 printed pages; Volume 2: 33 printed pages plus a 9-

Please consider and let us know:

- When would you be able to submit a complete manuscript?
- Are there any events or conferences where we can market this book?
- Would be appropriate to publish in view of factors such as regulatory updates, medical advances, etc.?

page artifact appendix). Typesetting-ready LaTeX sources with machine-checked cross-references and bibliographies can be submitted within 2–4 weeks of a positive decision.

Events and conferences for marketing: Principles and Practice of Constraint Programming (CP), the SAT Conference (both with artifact evaluation tracks suited to Volume 2), ICLP and MFCS for the theory audience; IMAC and the European Workshop on Structural Health Monitoring (EWSHM) for the engineering audience; the Computability and Complexity in Analysis conference and Weihrauch-complexity workshops for the computability tier.

Timeliness: the subject is timely rather than perishable — certified and independently checkable computation, the limits of agents acting on partial information, and reproducibility of computational studies are active growth areas; no regulatory or medical currency issues.

Specifications

Please address each of the following with your best estimates:

- Number of double-spaced Microsoft Word pages you anticipate producing **OR** the total number of words
- The approximate number of illustrations and figures you anticipate including, and whether any of these would require color
- What other **multimedia content** (audio/video files), maps, etc. can you include to increase the market value of the book?

Length. Volume 1: approximately 40,000 words (about 100 printed pages at research-monograph density; roughly 160–180 double-spaced Word pages). Volume 2: approximately 15,200 words including the artifact appendix (about 42 printed pages; roughly 60–65 double-spaced Word pages).

Illustrations and figures. Volume 1: 4 figures and 7 tables (notation table, diagnostic tables, obstruction tables). Volume 2: 10 tables (classification and benchmark tables); the artifact appendix references artifact-generated plots (certificate size versus width; verification time versus certificate size). All figures and tables are designed to be colour-independent — no colour required.

Multimedia and other content. The accompanying software artifact is the market differentiator: a public, version-frozen prototype implementing the certificate layer, the benchmark harness, and the recorded solver-comparison study, reproducible with one command (access details under Additional requirements). No audio or video content is planned.

Reviewers Suggested reviewers (contact details to be completed by the author; affiliations shown to indicate the disciplinary spread):

We use external reviews to validate the market for the book idea and provide suggestions that may help in its development. Please suggest at least experts in the field whose input you would value as part of our review panel, please provide their contact details below:

Name: Prof. Rina Dechter (University of California, Irvine) — constraint processing and graphical models

Email address: [email to be completed by author]

Name: Prof. Samson Abramsky

Email address: [email to be completed by author]

(University of Oxford) — sheaf-theoretic contextuality and categorical methods	
Name: Prof. Keith Worden (University of Sheffield) — structural health monitoring	Email address: [email to be completed by author]

Additional requirements

We can make a publishing decision based on a detailed contents list and full answers to the questions above. However, sample material is helpful in enabling reviewers to evaluate content and approach. Please provide a draft chapter or other sample content if available.

Sample material: complete, compiled manuscripts of both volumes (PDF; approximately 101 pages and 33+9 pages) are available immediately for review, together with the full LaTeX sources. Representative sample chapters can be designated on request; suggested samples are Volume 1's finite-gluing and contextuality chapter, and Volume 2's width-based certifying dynamic-programming chapter together with the artifact appendix.

Accompanying software artifact — access details:

– Repository: <https://github.com/MIKEAA2020/halting-physics> (public; no access request needed). The artifact lives in the `finite_ecd_prototype` directory; the exact version supporting the companion's computational study is frozen at release tag `halting_physics_workspace_v4` (frozen 2026-09-22). The repository also carries the full version history of both monographs and all implementation reports.

– Contents: a dependency-free Python prototype implementing the certificate layer (instance loading and well-formedness checks; policy verification; brute-force solving; fibre-conflict, sensor-repair, and sparsification certificates; the tree-decomposition dynamic program with typed certificates; finite-element row generation); the benchmark generator and sweep harness; the external-solver comparison layer (MiniSAT 2.2, Kissat 4.0.4, Z3 5.1.0, OR-Tools CP-SAT 9.15, HiGHS 1.15.1) with the cross-checking discipline; and the recorded JSON reports behind every table in the computational chapters.

– Reproduction: "make reproduce" (or explicit per-step scripts) under the recorded environment — Python 3.12 on standard x86-64 Linux; the recorded experiments ran on a 2-core, 4 GiB virtual machine, so no special hardware is needed. A release manifest (`RELEASE_MANIFEST.md`) pins the environment, solver versions, and freeze date; a version-exact replication of the recorded comparative study is part of the recorded runs.

– Persistence: a DOI or archival identifier (e.g., Zenodo) is planned to be assigned to the frozen release before publication, per the release manifest.

Publishing ethics and Generative AI (GAI)

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the review process.

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